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Research Article
doi.org/10.1002/cssc.202300106

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An Integrated Plasma–Photocatalytic System for Upcycling of Polyolefin Plastics

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Nondegradable polyolefin plastics, which account for >60% of total plastic waste, trigger severe global concerns and thus demand effective management technologies. However, owing to the chemical inertness of non-polar C–C backbones in the polyolefin structure, efficient upcycling of polyolefin plastics under ambient conditions remains a great challenge. This study introduces an integrated plasma–photocatalytic technology, coupling plasma treatment with solar-driven reforming under mild conditions, for the efficient upcycling of polyolefin plastics into value-added hydrogen and gaseous fuels. The first plasma step grafts oxygenated groups, such as –OH, O–C=O, and C=O, onto the polyolefin chains, which leads to the formation of a polar and hydrophilic polymer that facilitates the subsequent reforming in the photocatalytic step. Therefore, high hydrogen production activity with a benchmark efficiency of >100 μmol g^{-1} h^{-1} was achieved. Moreover, the integrated process also demonstrates high versatility in upcycling different polyolefin plastics including polyethylene, polypropylene and polyvinyl chloride. The findings provide a new avenue for plastic upcycling in an efficient and sustainable way.

Introduction

Plastics have been used worldwide due to their convenience, long durability, and low cost. Global plastic production has reached 367 million tons in 2020 and the demand for plastics surged due to the epidemic prevention requirements since the outbreak of COVID-19, putting greater pressure on plastic management.[1–3] Plastic waste and derived microplastics are invading the environment and even threatening human health, which has become a global issue of widespread concern.[4–6] To date, great efforts have been made for the treatment of plastic wastes,[7] including conventional methods such as landfill, incineration, and mechanical recycling,[8] as well as advanced methods that are being developed such as pyrolysis,[9,10] liquefaction,[11,12] gasification,[13,14] dissolution-based approaches,[15,16] biocatalysis,[17,18] and photocatalysis.[19–21]

Upcycling of plastics is a promising route for plastic valorization as it can upgrade the plastic waste into valuable fuels, chemicals, or materials with additional value.[22–25] Compared with thermal catalytic process, which is usually conducted under harsh conditions (high temperature and high pressure) and requires large energy input,[26,27] photocatalytic process provides a sustainable approach to utilize solar-energy for plastic conversion under ambient conditions. Moreover, the mild conditions of photocatalysis have great potential for the precise activation of designated chemical bonds and selectively produce targeted products. Specifically, photoreforming of plastics emerged as an effective photocatalytic process for plastic waste valorization accompanied with clean H_2 production.[28] In the photoreforming process, the photoexcited holes oxidize plastic waste into valuable chemicals, and meanwhile, photogenerated electrons reduce the protons in H_2O to H_2. In 1981, Kawai and Sakata first reported the photoreforming of plastics into H_2 using Pt|TiO_2 photocatalyst under UV light.[29] Recently, Uekert et al. developed CdS/CdO_x quantum dot for efficient plastic valorization under visible light, followed by other works that implemented various photocatalysts including CN_x|Ni_2P, CPDs-CN (carbonized polymer dots-graphitic carbon nitride), and MXene/$Zn_xCd_{1-x}S$ to expand the application on the photoreforming of plastics.[30–37]

However, it is noteworthy that most studies of plastic photoreforming focused on the conversion of polar polyester plastics such as polyethylene terephthalate (PET) and polylactic acid (PLA). These plastics can be hydrolyzed to corresponding monomers like ethylene glycol and sodium lactate in strong alkaline solution that can be chemically valorized through subsequent photoreforming.[37] In contrast, non-polar polyolefin plastics such as polyethylene (PE), polypropylene (PP), and polyvinyl chloride (PVC), accounting for >60% of plastic wastes,[38] have rarely been upcycled through photoreforming process. Compared with polyesters, polyolefins have low polarity and high stability due to the inert C–C and C–H bonds, which is difficult to be activated and upgraded. Furthermore, since photoreforming is an aqueous process and requires intimate interaction between plastics and photocatalysts, the low polarity of polyolefins impedes its dispersion in solution and decrease the interaction with catalysts, which lead to low H_2 evolution activity.[39,40] Therefore, developing an effective approach to upcycle polyolefin plastics under ambient conditions remains a great challenge.

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Supporting information for this article is available on the WWW under https://doi.org/10.1002/cssc.202300106

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Herein, we address the challenge by designing an integrated plasma–photocatalytic system that couples plasma treatment with photoreforming. The polyolefin plastic was first treated by non-thermal plasma to graft oxygenated groups for forming polar polymer, which facilitated the subsequent reforming in the photocatalytic step (Figure 1). The integrated tandem process also demonstrated high versatility in upcycling different polyolefin wastes and produce hydrogen at a benchmarking activity over 100 $\mu mol\, g^{-1} h^{-1}$. This work sheds light on the establishment of a new paradigm for plastic upcycling in an efficient and sustainable way.

Results and Discussion

Characterization of plasma-treated PE

Firstly, plasma treatment was applied for the pre-oxidation of plastics. During plasma processing, an electrical field was applied between two parallel electrodes. The air filled in the reactor will be ionized, excited, and dissociated by the electric field and create new radicals through the dissociation collisions of ions, electrons, and ultraviolet photons,[41] leading to the bond breaking and recombination of polymers. PE is one of the most widely used plastics, which accounts for about 35% of waste plastics, thus PE was selected as the model polyolefin. Sampling points were taken at ten-minute intervals during plasma treatment and the pristine and resulting treated PE were denoted as PE-x ($x = 0$, 10, 20, 30, 40, 50, and 60, respectively). Comprehensive characterizations of PE were carried out to have a better understanding of the plasma effect on the plastic features and corresponding photoreforming activity.

During the plasma processing, the high-energy plasma cleaves the C–C or C–H bond and grafts oxygenated groups onto the polymer backbones. The grafted functional groups had no significant effect on the molecular weight (see the Supporting Information, Figure S1) but impacted the surface properties of PE. FTIR was used to analyze the variations in the functional groups of PE surface during plasma treatment (Figure 2a). PE mainly consists of hydrocarbon chains and branches, thus its FTIR spectrum showed vibrations of C–H stretching in the range of 3100 cm^{-1} to 2700 cm^{-1}, C–H bending at 1465 cm^{-1} and rocking mode of the methylene group at 720 cm^{-1}.[42] After plasma treatment, absorption peaks assigned to oxygenated groups significantly enhanced. The peaks at 3440 cm^{-1} and 1260 cm^{-1} increased with the extension of plasma treatment time, which was attributed to the –OH stretching and C–O stretching vibrations, respectively (Figure S2a).[43] With plasma treatment, a peak at 1640 cm^{-1} also arose due to the C=O stretching of ketones, aldehydes, and carboxylic acids (Figure S2b).[44,45] Therefore, it can be concluded that the plasma treatment caused the cleavage of C–H bonds and formation of oxygenated groups including –OH, O–C=O, and C=O on PE surface.

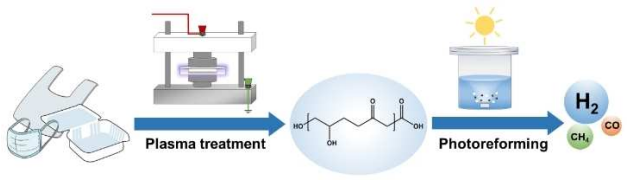

Figure 1. Schematic illustration of plastic conversion by plasma–photocatalytic process.

The oxidation degree of PE was quantitatively analyzed by calculating the carbonyl index (CI) and hydroxy index (HI).[46] The CI and HI were calculated through FTIR as previously reported,[47] and the baseline was adjusted with the PE-0 not containing functional groups. The CI was calculated as $CI = A_{C=O}/A_{1468}\text{-}CI_0$, where $A_{C=O}$, A_{1468}, and CI_0 are peak areas of carbonyl groups (1640 cm^{-1}), internal reference, and original CI before treatment, respectively. The HI was calculated as $HI = A_{OH}/A_{1468}\text{-}HI_0$, where A_{OH}, A_{1468}, and HI_0 are peak areas of hydroxy groups (3440 cm^{-1}), internal reference, and original HI before treatment, respectively. HI increased steadily with the extension of treatment time (Figure 2b), indicating that plasma ionized oxygen and formed hydroxy groups on PE surface continuously. Compared with HI, the growth rate of CI was lower before 30 min. But CI rose rapidly for PE-40, PE-50, and PE-60, suggesting that prolonged plasma treatment resulted in formation spurt of highly oxidized carbonyl or carboxyl groups on PE surface. As previous literature reported, the color change of plastic was caused by the increase of chromophore groups such as conjugated carbonyl groups.[48] After plasma treatment, PE-60 showed a faint yellow color compared with white color of PE-0, indicating that PE was oxidized obviously by plasma after 60 min.

To investigate the functional group changes on the PE surface, XPS analysis was performed. From the full scanned XPS spectra (Figure 2c) and elemental atomic percentages (Table S1), it can be seen that the oxygen content increased from 0.76% of PE-0 to 3.52% of PE-60 continuously on account of the grafting of oxygen-containing groups on PE surface. From the O1s fine structure spectra (Figure 2d), O1s signals can be divided into two peaks with binding energies of around 532.3 and 533.2 eV, attributed to oxygen of C=O and C–O, respectively. It was found that the peak areas of the two peaks increased significantly, indicating that the contents of C–O and C=O bonds both increased in terms of plasma treatment time. The XPS results proved that plasma treatment increased the content of –OH, O–C=O, and C=O functionalities on PE surface and long-time plasma treatment led to excessive oxidation of PE.

The microstructure and specific surface area of PE before and after plasma treatment was also investigated by SEM and BET analysis. Figure S3 shows the SEM images of PE with different treatment times. It was observed that there was no obvious difference in the roughness and fragmentation degree of PE particles. Therefore, the surface morphology of PE was not greatly affected by plasma treatment due to the low discharge power. Additionally, an infrared imager (FLIR A700) was employed to monitor the temperature during plasma treat-

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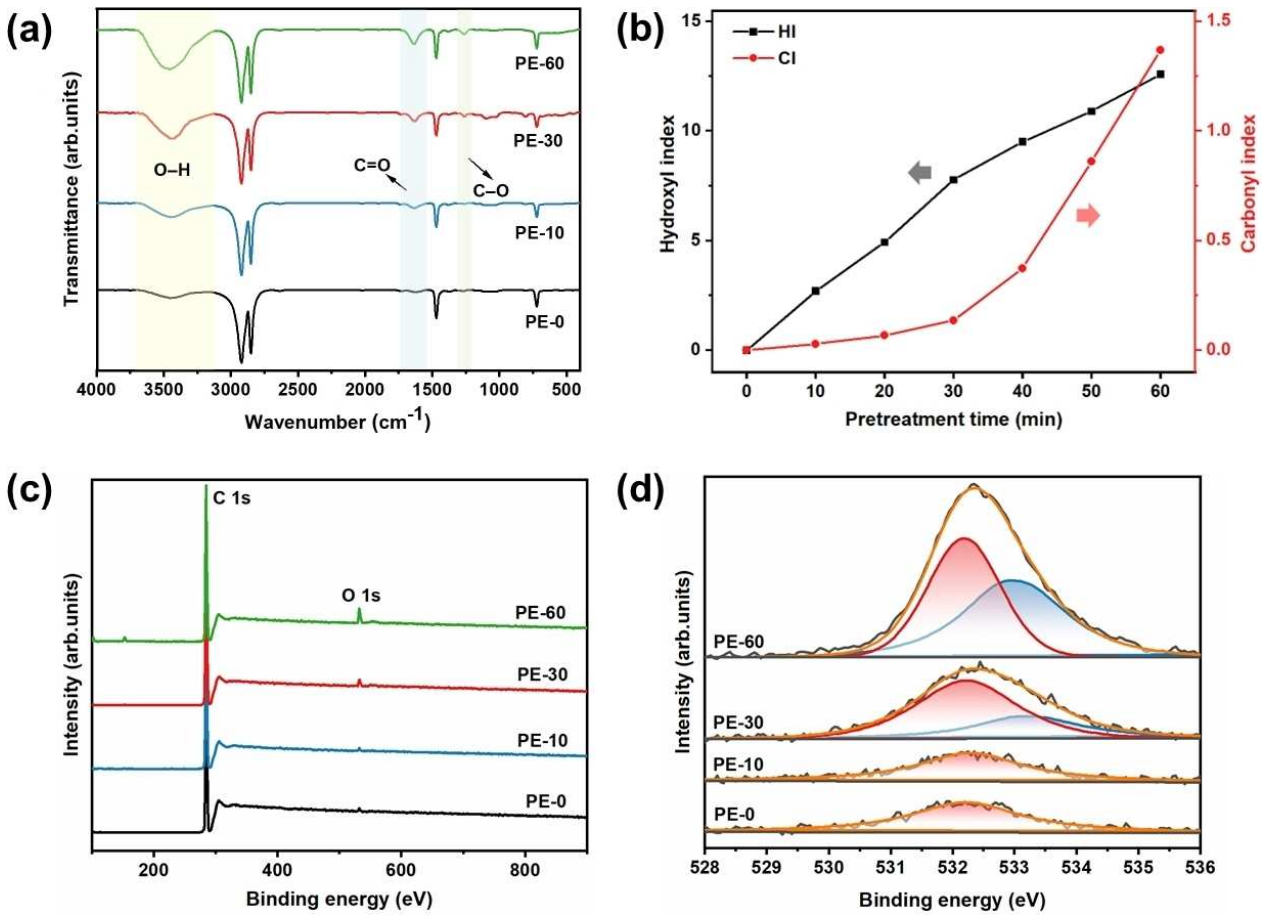

Figure 2. Chemical structure analysis of PE with different plasma treatment times: (a) FTIR spectra; (b) hydroxy and carbonyl index evolution; (c) full scanned XPS spectra; (d) O1s fine structure spectra.

ment. As shown in Figure S4, the temperature of plasma reactor increased slightly and the final temperature after 60 min was about 24.1 °C, which will not lead to the deformation and melting of PE. The specific surface area of PE powder was characterized by the N_2 adsorption-desorption test. The specific surface areas of PE-0, PE-10, PE-30, and PE-60 samples were 0.26, 0.27, 0.26, and 0.23 $m^2 g^{-1}$, respectively. The results illustrated that the microstructure of PE was not affected by plasma treatment.

Polarity of plasma-treated PE

Owing to the low polarity of pristine PE, it is difficult for PE to dissolve in polar solvents such as H_2O, which limits the plastic-catalyst interaction. With the plasma treatment, the polarity of PE is changed, leading to the improvement of hydrophilicity and enhancement of interaction between plastic and photocatalyst. Consequently, the subsequent photocatalytic reactivity of plastics is promoted on account of the polarity switch.

Hydrophilicity is an important index to measure molecular polarity. When the same mass of PE powder was added to deionized water and stirred for the same time, the suspension presented different states for PE with different treatment times (Figure 3). The pristine PE powder floated intact above the water, whereas the treated PE powder was uniformly dissolved and suspended in the solution. The longer the treatment time, the less PE powder floated above the water. As a function of polarity, water contact angle is a macroscopic property that quantifies the hydrophilicity of materials, thus the smaller the contact angle of PE, the better hydrophilicity it has.[49,50] The contact angle of PE powder decreased from 140.2° to 78.0° as a function of plasma treatment time (Table S1), which indicated the improvement of hydrophilicity. With the hydrophilicity enhancement, plastic particles obtain larger surface free energy and can disperse better in aqueous solution.

The polarity improvement will also promote the adsorption between polymer and catalyst.[51] Computational methods have proven to be a powerful tool in explaining the adsorption mechanism and in achieving a profound understanding of adsorbate-adsorbent interactions. It was mentioned in the previous reports that the conversion of plastic took place through the interaction between plastic and semiconductor,[52] thus the molecular-level simulation was conducted by Material Studio in this study to further explore the adsorption between PE and TiO_2 before and after plasma treatment.

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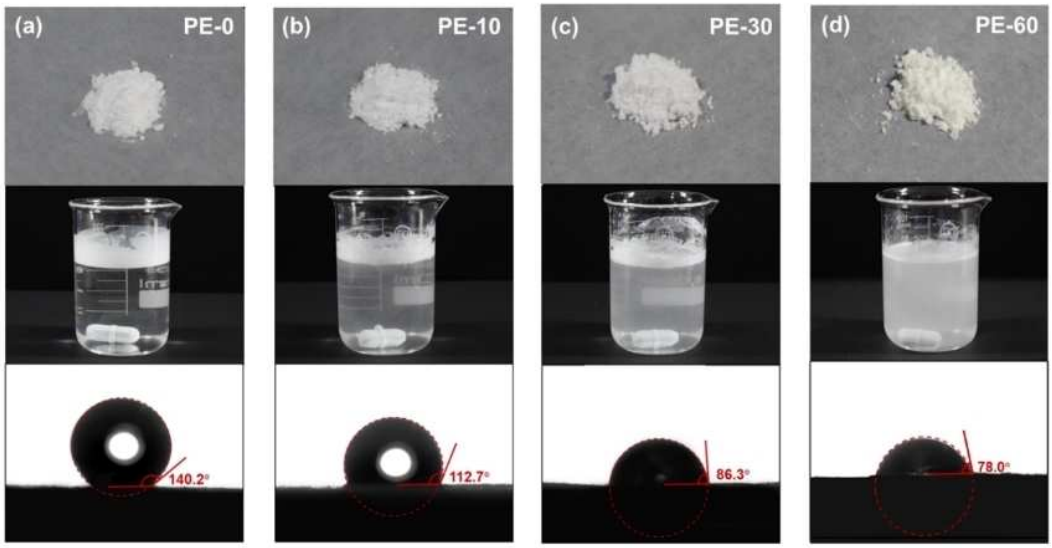

Figure 3. Photographs of PE samples with different plasma treatment times dispersed in water and corresponding water contact angle measurements.

A polymer consisting of 15 ethylene monomers was constructed as PE model. To simulate the plasma-treated PE, functionalities like –OH, O–C=O, and C=O were randomly inserted into the carbon chains of the PE model. From the simulation results (Figure 4), we found that the PE was in closer contact with TiO_2 surface after plasma treatment. The interaction energy gave insights into the stability of the adsorbate's molecule on the adsorbent's surface. The calculated interaction energies were found to be -29.85 kcal mol^{-1} (pristine PE) and -45.04 kcal mol^{-1} (plasma-treated PE). The higher negative value indicates that the treated PE powder showed stronger interaction with the photocatalyst through collision, leading to closer contact between surface active sites and PE particles.

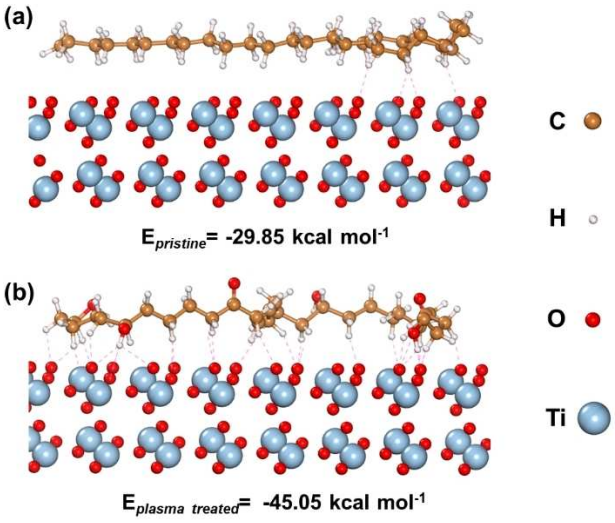

Figure 4. Illustration of the interaction between polyethylene and TiO_2 surface before and after plasma treatment.

Photoreforming performance

Eventually, the purpose of plasma treatment was to improve the reactivity of PE through the change of polarity. The effect of plasma treatment on H_2 production activity in photoreforming process was studied. Commercial catalyst P25-TiO_2 was used as the model photocatalyst for plastic photoreforming. Pt nanoparticles with sizes of 5–15 nm were deposited on TiO_2 as co-catalyst for promoting H_2 production (Figure S5). The H_2 evolution rates of different samples were compared (Figure 5a). The untreated PE-0 showed the lowest activity, with H_2 evolution rate of only 33.22 μmol g^{-1} h^{-1}. With the extension of plasma treatment time, the H_2 evolution rate gradually rose to 58.46 and 73.36 μmol g^{-1} h^{-1} for PE-10 and PE-20, respectively (Table S2). As the treatment time was extended to 30 min, the H_2 evolution rate further rose to 108.95 μmol g^{-1} h^{-1}, which was 3.3 times higher than that of the pristine PE. However, the photoreforming activity decreased when plasma treatment time exceeded 30 min, and the longer the treatment time, the more severe the decrease in activity. The H_2 evolution rate even went down to the same level of pristine PE for PE-60, indicating that H_2 evolution rate did not increase indefinitely with the plasma treatment time.

Combined with the experimental observations and theoretical calculations, it can be explained that the grafting of oxygen-containing groups on PE surface leads to better hydrophilicity, and stronger interaction between PE and photocatalyst, resulting in a more efficient H_2 production. During the preliminary plasma treatment process, the increase of hydroxy groups is the predominant factor contributing to the enhanced reactivity. As reported in previous reports,[53,54] a linear relationship between the number of hydroxy groups and quantity of generated H_2 was found, proving the positive effect of the grafted hydroxy groups on photoreforming process. But long-time plasma treatment will give rise to excessive oxidation of

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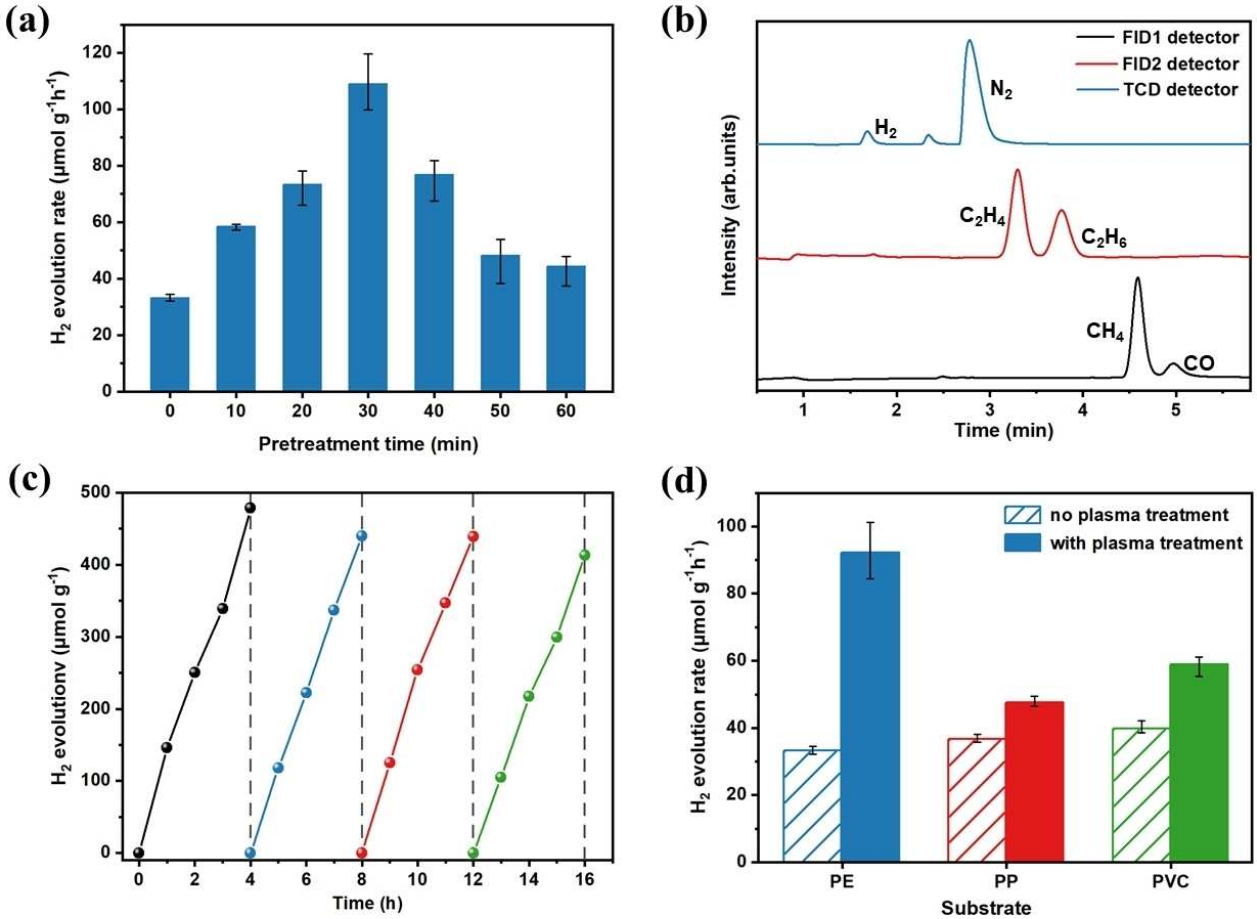

Figure 5. (a) H_2 evolution rate of PE with different treatment times. (b) GC spectra of gas product after 24 h photoreforming. (c) Cycling tests of Pt|TiO_2. (d) H_2 evolution rate of different plastics before and after 30 min plasma treatment. The error bars represent one standard deviation of three independent measurements.

PE, generating large amounts of carbonyl and carboxyl groups, which was the main reason for the decrease of H_2 production activity. As shown in Figure S6, carboxylic acids have lower H_2 evolution rate as sacrificial agents compared with polyols in photoreforming process, which is consistent with a previous report.[55] It was also shown in the previous study that formic acid would inhibit the performance of Pt|TiO_2 photocatalyst and cause the deactivation problem.[56] In terms of performance, it is noteworthy that the H_2 evolution rate of PE-30 (108.95 $\mu mol\, g^{-1} h^{-1}$) is higher than that reported in an integrated chemical-photocatalytic process using the same catalyst (65.63 $\mu mol\, g^{-1} h^{-1}$), which couples nitric acid hydrothermal treatment and photoreforming process.[57,58] In addition, the current plasma treatment process is considered to be more environment-friendly than the harsh acid treatment from the viewpoint of green chemistry.

In the photoreforming process, H_2 was generated by reduction half-reaction of H_2O, and the plastics were oxidized to valuable products by oxidation half-reaction. Under anoxic conditions, PE was found to be converted to hydrocarbons including CH_4, C_2H_4, and C_2H_6 (Figure 5b), as TiO_2 had the ability to transfer the adsorbed hydrogen to the intermediate radical effectively,[57] and CO_2 was the main byproduct. There was no great difference in product distribution between pristine and treated PE as shown in Table S3. It is worth mentioning that a higher proportion of ethane was detected for treated PE, which indicates that plasma treatment accelerates the process of photoreforming, promoting the hydrogen transfer reaction. No significant accumulation of liquid products was detected by ^{1}H NMR either in short-term or long-term photocatalysis tests (Figure S7). The conversion of plastics into gaseous products facilitates the easier separation of products. In addition, the stability of photocatalyst was also measured by four-time consecutive H_2 evolution tests. During four cycled tests with a cumulative irradiation time of 16 h, the catalyst showed stable performance with an average H_2 production rate of 103.21 $\mu mol\, g^{-1} h^{-1}$ (Figure 5c).

Generalization to other polyolefin plastics

The application of plasma–photocatalytic system was extended to other polyolefin plastics, including PP and PVC, which accounting for 19% and 10% of the entire plastic industry,

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respectively.[7] Structurally, PP and PVC have similar backbone as PE, with hydrogen atom replaced by methyl group and chlorine atom, respectively. The H_2 evolution rates of different plastics during photoreforming before and after the plasma treatment are shown in Figure 5d. With 30 min plasma treatment, the photoreforming activity for PP and PVC were improved to be 56.32 and 69.64 $\mu mol\,g^{-1}\,h^{-1}$, respectively. However, the increasing ratio of the H_2 evolution rate for PP and PVC (1.5 and 1.8 times, respectively) is less pronounced than the 3.3-fold increase for PE. This is due to the slight increase of oxygenated groups such as –OH during plasma treatment for PP and PVC, as shown in the FTIR analysis (Figure S8). Slight dechlorination was also found for PVC from FTIR (Figure S9), indicating that the existence of side groups in PP and PVC may generates steric effect and impedes the formation of polar component as discussed in previous study.[59] Although quantitative differences exist between different plastics, similar qualitative trend indicates the high versatility of plasma–photocatalytic system for plastic valorization.

Conclusions

In summary, an integrated plasma–photocatalytic system was designed for the efficient upcycling of polyolefin plastics into hydrogen and gaseous fuels. The first plasma treatment graft oxygenated groups to form polar polyolefin structure, which facilitated the subsequent photoreforming step and produce hydrogen at a benchmark activity of over 100 $\mu mol\,g^{-1}\,h^{-1}$. The integrated process also demonstrated high versatility in upcycling different polyolefin wastes including PE, PP and PVC. This work may open new opportunities for the design and development of efficient systems for plastic waste valorization in a green and sustainable manner.

Experimental Section

Materials

HDPE, PP, PVC, and PET were purchased from Zhongxing Plastics Co., Ltd., China, and detailed information is given in Table S4. Methanol, formic acid, P25-TiO_2 and chloroplatinic acid were purchased from Shanghai Macklin Biochemical Co., Ltd., China. All chemicals were analytically pure and used as received without further purification.

Plasma treatment

The plasma treatment was conducted on a dielectric-barrier-discharge (DBD) plasma system (Nanjing Suman Electronics Co., Ltd., China). The system consisted of a power source with an output of 0–2000 W and a reactor composed of two parallel circular stainless steel plane electrodes and a quartz sample container between the electrodes. The discharge distance was exactly controlled by the quartz container, which consisted of a concave quartz plate (external diameter of 145 mm, internal diameter 60 mm, external depth of 10 mm, internal depth of 8 mm) and an upper quartz cover (diameter of 150 mm, thickness of 2 mm). 0.1 g plastic powder was placed in the reactor with great care to obtain a quasi-monolayer of almost constant thickness. With the monolayer, the diffuse DBD plasma could cover most surface of the powder.[60] The modification process was conducted under room temperature and atmospheric pressure in air flow at a rate of 100 $mL\,min^{-1}$. The voltage and current supplied to the reactor were monitored by a digital oscilloscope (Tektronix TBS 1102) with a 1:1000 high-voltage probe and a current probe. The peak voltage supplied to the reactor was 40 kV and the discharge voltage and current waveforms are shown in Figure S10. The discharge characteristic was consistent with the typical atmospheric gas-phase DBD.

Characterization of plastics

The pristine and plasma-treated PE powder was characterized by the Fourier transform infrared (FTIR) spectra on a Thermo Scientific Nicolet Is5 spectrometer, the X-ray photoelectron spectra (XPS) on a Thermo Scientific K-Alpha with an $Al_{K\alpha}$ (1486.6 eV) radiation source, the scanning electron microscopy (SEM) images on a Zeiss Supra 55 microscope, and the Brunauer–Emmett–Teller method (BET) on a Micromeritics ASAP 2460. The contact angle of PE powder was measured by KRUSS DSA100 automatic contact angle measurement. The weight-average molecular weight (M_w) of PE were analyzed by a PL-GPC220 gel permeation chromatography system.

Computational calculation

Molecular dynamics (MD) was carried out to study the adsorption between polyethylene and TiO_2 using Forcite, Amorphous as well as the Build layer modules implemented in Material Studio 8.0 program. The TiO_2 was imported from Materials Studio crystal structures and cleaved to produce anatase (101) surface as it covers most surface of P25-TiO_2 and forms the majority fraction for catalytic interfaces.[61] Considering that TiO_2 is an ionic compound, the Ti–O bond was deleted before surface relaxation. The uppermost two layers of the plane were relaxed to act as surface layers, while the innermost layers were kept fixed to be bulk layers. A polymer fragment containing 15 monomers was constructed by Amorphous cell module and then placed together with the optimized anatase (101) surface using the Build layer module. Functional groups include –OH, O–C=O, and C=O were inserted in the long chain of polymer model randomly to simulate the plasma-treated PE and placed together with the surface as well. The constructed simulation system was fully optimized and then submitted for MD simulations using Forcite module with the following details: COMPASSII force field, NVT ensemble, a time step of 1.0 fs, simulation time of 3000 ps, 303 K maintained constant by the Andersen thermostat. The interaction between PE and the photocatalyst surface was estimated by using Equation (1):[62]

$$E_{interaction} = E_{total}\text{-}(E_{surface} + E_{polymer}) \quad (1)$$

where E_{total}, $E_{surface}$, $E_{polymer}$ are the single-point energies for the whole system, the photocatalyst surface, and the polymer alone, respectively.

Photoreforming of plastic

Before photoreforming process, 1 wt% Pt was loaded on the photocatalyst as co-catalyst through photodeposition.[63,64] Briefly, 50 mg TiO_2 was dispersed in 30 mL H_2O with sonication for 10 min. Then, 125 μL H_2PtCl_6 solution (8 wt% in H_2O) and 3 mL methanol were added into the reactor. After further sonication of 10 min, the mixture was irradiated under a Xe lamp for 2 h with continuous

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stirring. After the deposition, the catalyst was washed with H_2O and dried at 80 °C overnight. The photocatalyst was characterized by the X-ray diffraction (XRD) on a Rigaku Ultima IV X-ray diffractometer with $Cu_{K\alpha}$ X-ray radiations and the transmission electron microscopy (TEM) on a FEI Talos 2005 transmission electron microscope.

The photoreforming process was conducted in a sealed quartz container filled with N_2. In a typical experiment, 30 mg Pt | TiO_2 and 30 mg plastic powder were dispersed in the 100 mL reactor containing 30 mL deionized water. After stirring for 30 min, the suspension was irradiated by simulated solar light (AM 1.5G, 100 mW cm^{-2}) at 25 °C. For recycling test, the gas in the reactor was replaced by N_2 without recovering the photocatalyst. The gas products were tested by gas chromatogram (GC-14C, Shimadzu, Ar carrier gas), and the filtrate after reaction was analyzed by ^{1}H NMR spectroscopy (Bruker Avance II300) with dimethyl sulfoxide (DMSO) as the internal standard.

Acknowledgements

This work was supported by the National Natural Science Foundation of China (22005048, 51822604), the Natural Science Foundation of Jiangsu Province (No. BK20200399), and the Scientific and Technological Innovation Project of Carbon Emission Peak and Carbon Neutrality of Jiangsu Province (No. BE2022028-4). We thank Prof. Daolun Liang for the help in infrared temperature measurement. Sheng Chu would like to thank the support from the "Zhishan Young Scholar" Program of Southeast University.

Conflict of Interests

There are no conflicts to declare.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords: hydrogen production · photocatalysis · photoreforming · plasma treatment · plastics upcycling

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Revised manuscript received: March 14, 2023
Accepted manuscript online: March 28, 2023
Version of record online: May 17, 2023

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